IEEE PROJECT REPORT

AI CHATBOT BASED TICKETING SYSTEM

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Abstract

AI Chatbot Based Ticketing System is an intelligent booking platform designed to help users search, compare, and reserve transport tickets through conversational interaction. The system integrates natural language processing and real-time ticket workflows to simplify booking and improve user experience.

Keywords

Artificial Intelligence, Chatbot, Ticket Booking, NLP, Web Application, Automation

Introduction

Traditional ticket booking platforms often require multiple screens and manual navigation. This project introduces an AI-powered chatbot interface to streamline booking, price checking, and user interaction.

Problem Statement

Users face delays and complexity while booking transport tickets. The proposed chatbot reduces interaction effort and improves accessibility.

Objectives

1. Build an AI chatbot interface.
2. Provide ticket search and booking assistance.
3. Improve user experience using conversational AI.

Methodology

Frontend: React/TypeScript
Backend: FastAPI/Python
AI Layer: LLM/NLP Integration
Database: PostgreSQL or MongoDB
APIs: Transport/Ticket integration

System Architecture

User → Chat Interface → Backend API → AI Engine → Booking Service → Database

Expected Outcomes

Faster booking process, improved accessibility, and intelligent customer interaction.

Future Scope

Voice-based booking, multilingual support, recommendation systems, and predictive pricing.

Conclusion

The AI chatbot-based ticketing system demonstrates how conversational AI can modernize ticket booking workflows.

References

[1] IEEE Paper Format Reference
[2] Research articles on AI Chatbots and Ticketing Systems